LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	33
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Measurements of lift and drag in wind tunnel
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Measurements of lift and drag in wind tunnel.
5. Specific Objectives	6.1.5 .1Trainee will be able to explain about wind tunnel operation

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of wind tunnel instrumentation setup, safety checks for high-speed airflow, and instructor supervision of data recording procedures.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a wind tunnel in aerodynamics?
(b) How does airflow velocity relate to lift generation?
(c) Why is drag measurement critical for aircraft performance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Wind tunnel operational principles • Lift measurement instrumentation • Drag measurement instrumentation • Data acquisition systems • Calibration procedures	L & Ds	PPT, IB	15 min
(b)	• Force balance mechanisms • Pressure tap distribution • Flow straightening grids • Boundary layer control • Scale model testing	L & Ds	PPT, PDF	20 min
(c)	• Interpretation of lift coefficients • Interpretation of drag coefficients • Error analysis in measurements • Reynolds number effects • Report generation standards	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a strain gauge measures lift force in a wind tunnel.
(b) Describe the role of a balance in separating lift and drag components.

12. Home Assignment and Follow Up	Prepare a brief note on how wind tunnel data informs the design of future aircraft wings for the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min